

Checklist: How to Reduce Bias in Your Data

AIM ASEAN Programme: AI for MSME Advancement in ASEAN

Curriculum Designed and Developed by: AIM ASEAN Programme

Confidentiality Notice:

This curriculum is specifically designed for the AIM ASEAN Programme and is intended solely for the training and development of MSMEs in the ASEAN region. Unauthorized use or reproduction of this material is prohibited.

For inquiries, please contact:
aimasean@aseanfoundation.org

Checklist: How to Reduce Bias in Your Data

Reducing bias in data is a continuous process, not a one-time fix. Use this checklist as a practical guide to identify and mitigate bias at every stage of your AI adoption journey.

Note that this list is a starting point and not meant to be exhaustive.

Before You Collect Data

[] **Define Your Problem Clearly:** Before gathering any data, ensure you have a precise and well-defined problem you want to solve. Ambiguous goals can lead to biased data collection.

[] **Form a Diverse Team:** Involve a diverse group of people (in terms of gender, age, background, and experience) in the planning stage. Different perspectives are crucial for spotting potential biases you may not be aware of.

[] **Identify Potential Sources of Bias:** Brainstorm where bias might creep into your data. Are you only collecting data from your most loyal customers? Are you relying on historical data that reflects past societal prejudices?

Other remarks:

During Data Collection

[] **Ensure Data Represents Your Target Population:** Your data should be a fair reflection of the people your AI will affect. If your customer base is diverse, but your data is mostly from one demographic, you have a representation bias.

[] **Check for Incomplete or Missing Data:** Understand *why* data might be missing. If certain demographic groups consistently have more missing data, it can skew your results. Don't just ignore it; investigate the cause.

[] **Use Multiple Data Sources:** Relying on a single source of data can amplify its inherent biases. Use a variety of data sources to get a more complete picture.

Other remarks:

After You've Collected Your Data

[] **Audit Your Data:** Systematically review your data for any signs of imbalance or skewed representation. Look for underrepresented groups or over-represented data points that don't match the real world.

[] **Look for Proxies for Protected Attributes:** Be cautious of using data points that might indirectly correlate with sensitive attributes like race, gender, or age (e.g., zip codes, type of car). AI can learn to use these as proxies to make biased decisions.

[] **Document Everything:** Keep a detailed record of where your data came from, how it was collected, and any changes or cleaning processes. This documentation is vital for transparency and helps you trace the source of any bias that may emerge.

Other remarks:

As You Use the AI Model

[] **Test for Fairness:** Before deploying your AI model, test it on different subgroups of your data to ensure it performs equally well for all of them. For example, a facial recognition tool should work just as well for women as it does for men.

[] **Implement Continuous Monitoring:** AI models can "drift" and become biased over time as new data is introduced. Set up a system to regularly monitor the AI's performance for fairness and accuracy, and have a plan to retrain it if needed.

[] **Get Feedback from End-Users:** Listen to your employees and customers. If they report that the AI is acting unfairly or producing strange results, take their feedback seriously and use it to improve your model and data.

Other remarks:
